

CSA 6503 - Generative Ai

CO2-AT3

25. Multi-Function Customer Service Agent – Function Calling Assessment Tool

A **Multi-Function Customer Service Agent** uses **function calling** to perform different customer support tasks by connecting an LLM to external functions or APIs.

Objective

Build an AI customer service agent that can:

- Check order status
- Cancel an order
- Book appointments
- Answer FAQs
- Escalate issues to a human agent

Workflow

Customer

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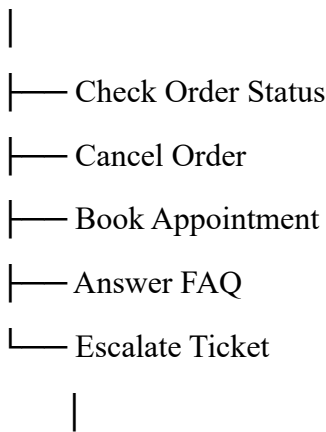


User Query

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Large Language Model (LLM)



Function Calling



Database / APIs



Response to Customer

Example Functions

Function	Purpose
check_order(order_id)	Returns order status
cancel_order(order_id)	Cancels an existing order
book_appointment(date, time)	Schedules an appointment
faq(question)	Answers common questions
create_ticket(issue)	Creates a support ticket

Sample Python Code

```
def check_order(order_id):  
    return f"Order {order_id} is Out for Delivery."  
  
def cancel_order(order_id):  
    return f"Order {order_id} has been cancelled."  
  
def book_appointment(date, time):  
    return f"Appointment booked on {date} at {time}."  
  
def faq(question):  
    faqs = {  
        "return policy": "Returns are accepted within 30 days.",  
        "shipping": "Shipping takes 3-5 business days."  
    }  
    return faqs.get(question.lower(), "Sorry, I don't know the answer.")  
  
def create_ticket(issue):  
    return f"Support ticket created for: {issue}"  
  
while True:  
    print("\n1. Check Order")  
    print("2. Cancel Order")  
    print("3. Book Appointment")  
    print("4. FAQ")  
    print("5. Create Ticket")
```

```
print("6. Exit")
```

```
choice = input("Enter choice: ")
```

```
if choice == "1":
```

```
    order = input("Order ID: ")
```

```
    print(check_order(order))
```

```
elif choice == "2":
```

```
    order = input("Order ID: ")
```

```
    print(cancel_order(order))
```

```
elif choice == "3":
```

```
    date = input("Date: ")
```

```
    time = input("Time: ")
```

```
    print(book_appointment(date, time))
```

```
elif choice == "4":
```

```
    q = input("Question: ")
```

```
    print(faq(q))
```

```
elif choice == "5":
```

```
    issue = input("Describe issue: ")
```

```
    print(create_ticket(issue))
```

```
elif choice == "6":
```

```
    print("Thank you!")
```

```
break
```

```
else:
```

```
    print("Invalid choice.")
```

Sample Output

1. Check Order
2. Cancel Order
3. Book Appointment
4. FAQ
5. Create Ticket
6. Exit

Enter choice: 1

Order ID: 12345

Order 12345 is Out for Delivery.

Enter choice: 4

Question: return policy

Returns are accepted within 30 days.

Enter choice: 5

Describe issue: Payment failed

Support ticket created for: Payment failed

Assessment Criteria

Criteria	Description
Function Selection	Chooses the correct function based on the user's request
Parameter Extraction	Correctly identifies inputs (e.g., order ID, date, time)
Response Accuracy	Returns the correct result from the function
Error Handling	Handles invalid or missing inputs gracefully
User Experience	Provides clear and helpful responses